

Direction Over Magnitude

Continuing from *Ensemble-Gated SpO2 Correction*: does the architectural residual bound itself explain the subject-shift failures the monitor was built to catch, or is the failure something a bound can't fix? Two experiments, both run without adding any new gating mechanism: a full retraining sweep across six values of `max_residual`, and a direct comparison of the learned correction against the mathematically ideal one.

File `train4.py` **Method** 11-fold LOSO, repeated per bound **Grid** `max_residual` $\in \{0,1,2,3,4,5\}$
Date 2026-09-04

WHERE THE LAST REPORT LEFT OFF

The prior report established that a leakage-free ensemble-disagreement signal (`cross_model_std_delta`) could predict when the learned SpO2 correction's *direction* was unreliable, and that gating on it produced a real but modest, uneven improvement over the uncontrolled model. This report asks two follow-up questions directly: is that unreliability actually a symptom of the correction being allowed too much or too little authority (the `max_residual` bound) — and, independent of any bound, is the learned correction even pointed the right way to begin with?

Sweeping the architectural bound

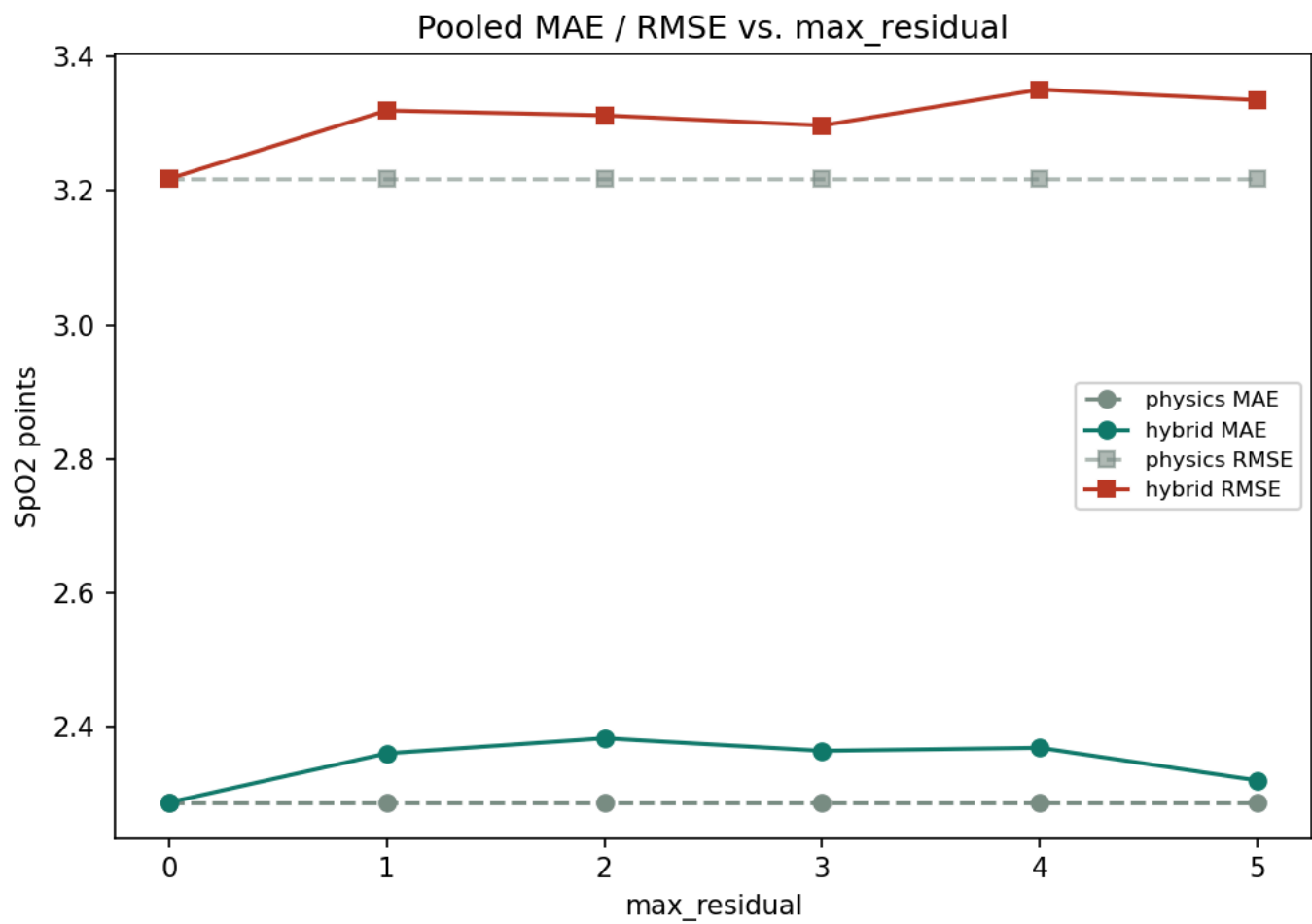
The full 11-fold LOSO protocol was repeated at six values of `max_residual` , holding every other choice identical — splits, seed, optimizer, epochs, architecture, physics calibration.

`max_residual=0` required no training at all: with the bound at zero, `delta = max_residual · tanh(raw_delta)` is identically zero, which mathematically zeroes the gradient to the entire learned pathway — verified, not assumed (`y_hat == y_linear` for every window).

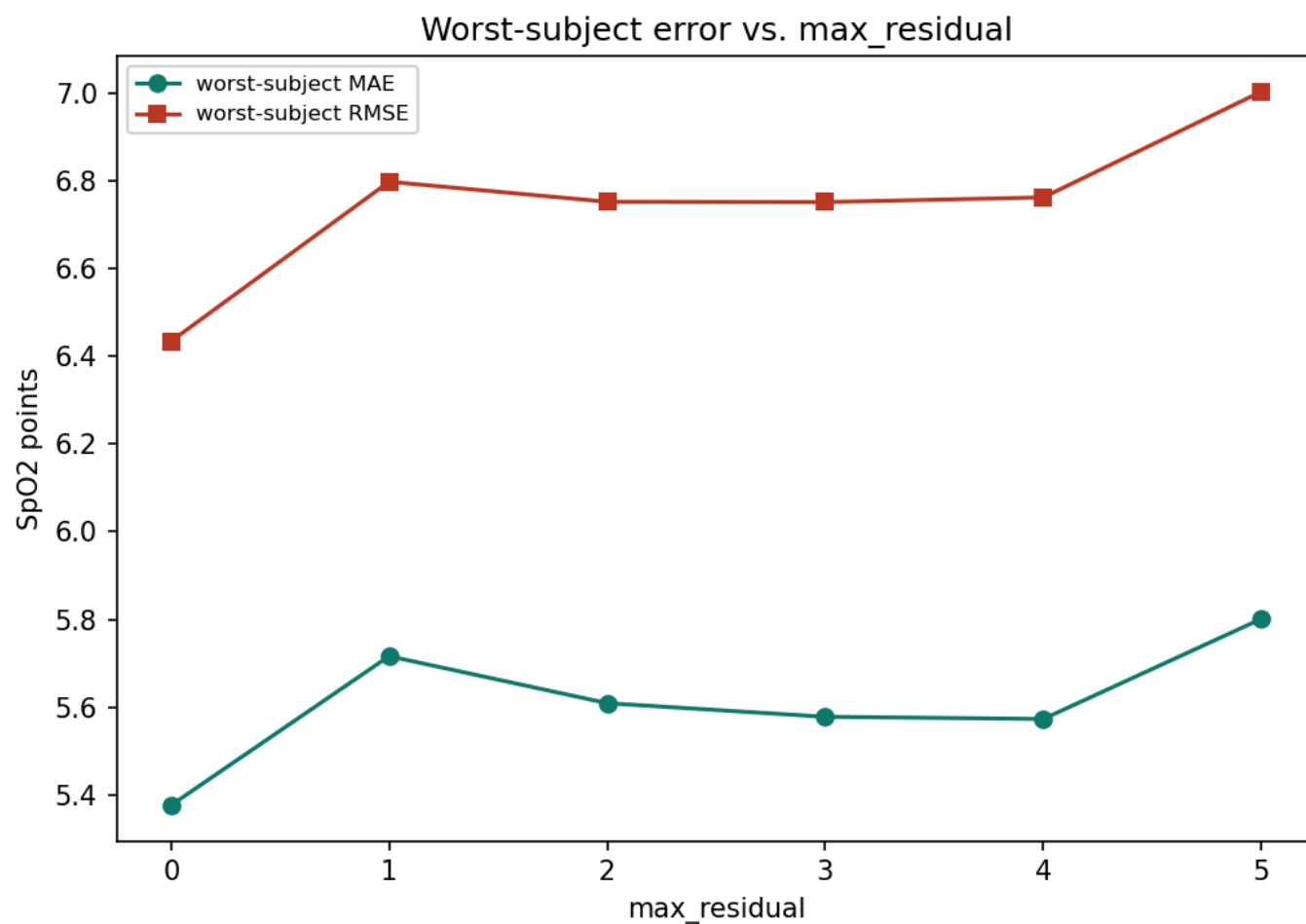
`max_residual=2` , the bound used throughout the rest of this project, reused the existing checkpoints rather than retraining a redundant copy. The remaining four bounds were genuinely retrained.

POOLED RESULTS BY BOUND

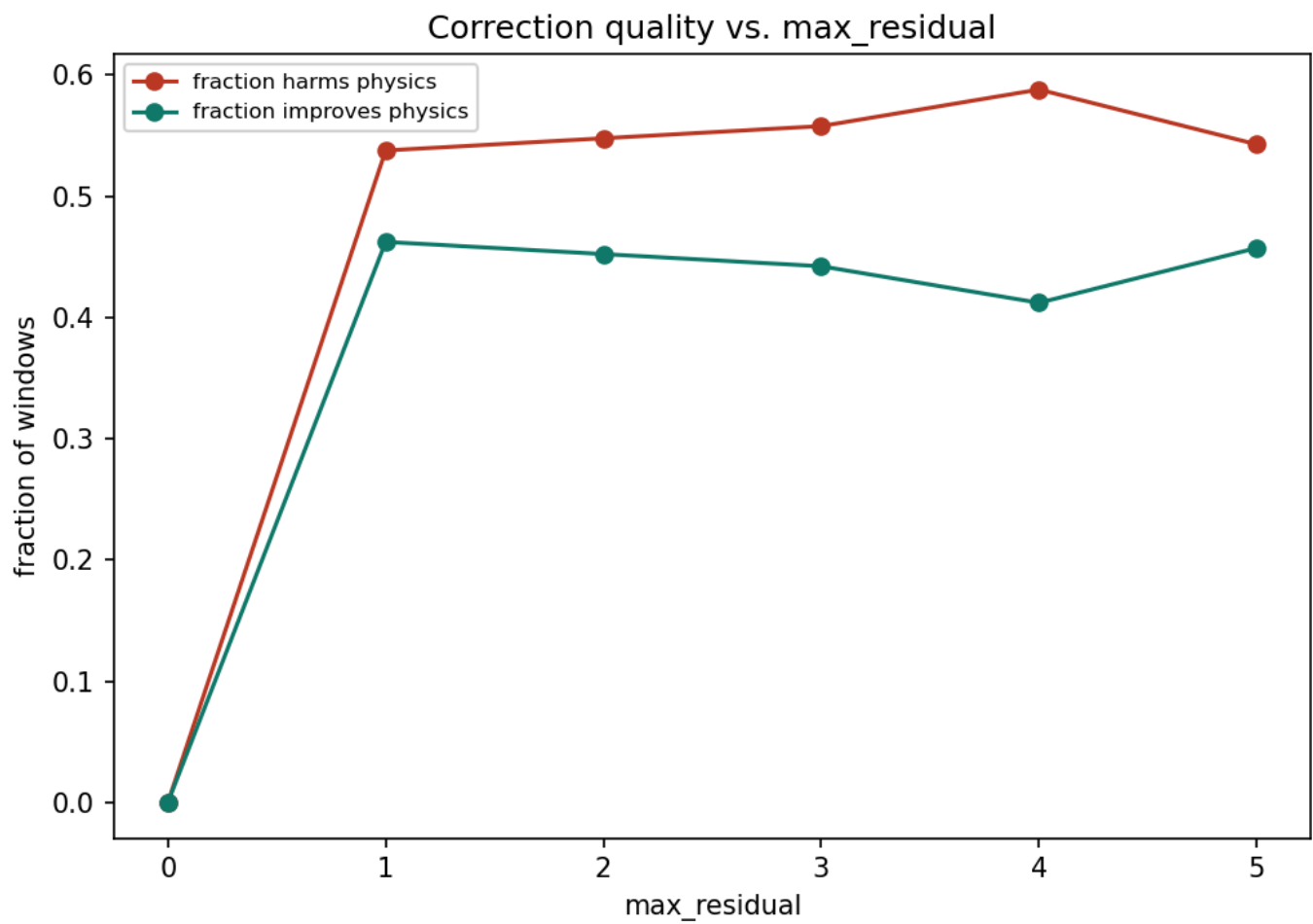
MR	HYBRID MAE	HYBRID RMSE	SAT. (P≥0.8)	SUBJ ↑/↓	WORST-SUBJ MAE	MONITOR GAIN
0 (physics)	2.287	3.218	0.0%	0/0	5.377	+0.000
1	2.360	3.320	29.1%	4/7	5.716	+0.068
2 (current)	2.383	3.312	7.0%	5/6	5.609	+0.092
3	2.364	3.297	2.5%	6/5	5.578	+0.112
4	2.369	3.351	1.5%	3/8	5.573	−0.005
5	2.320	3.335	1.0%	5/6	5.801	+0.092



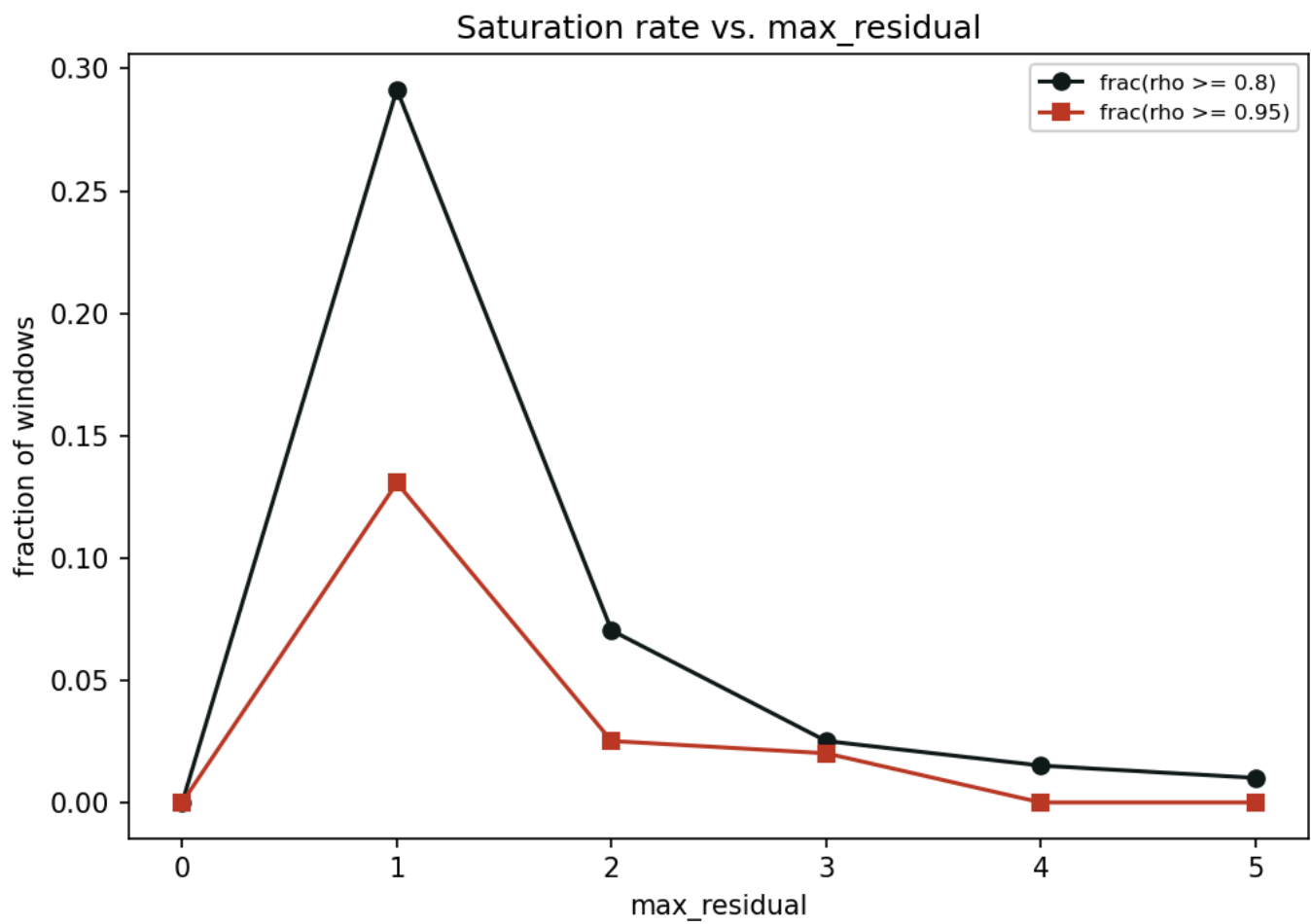
Pooled accuracy barely moves across mr=1–5 — the specific bound value matters far less than whether the network has any authority at all (the 0 → nonzero jump).



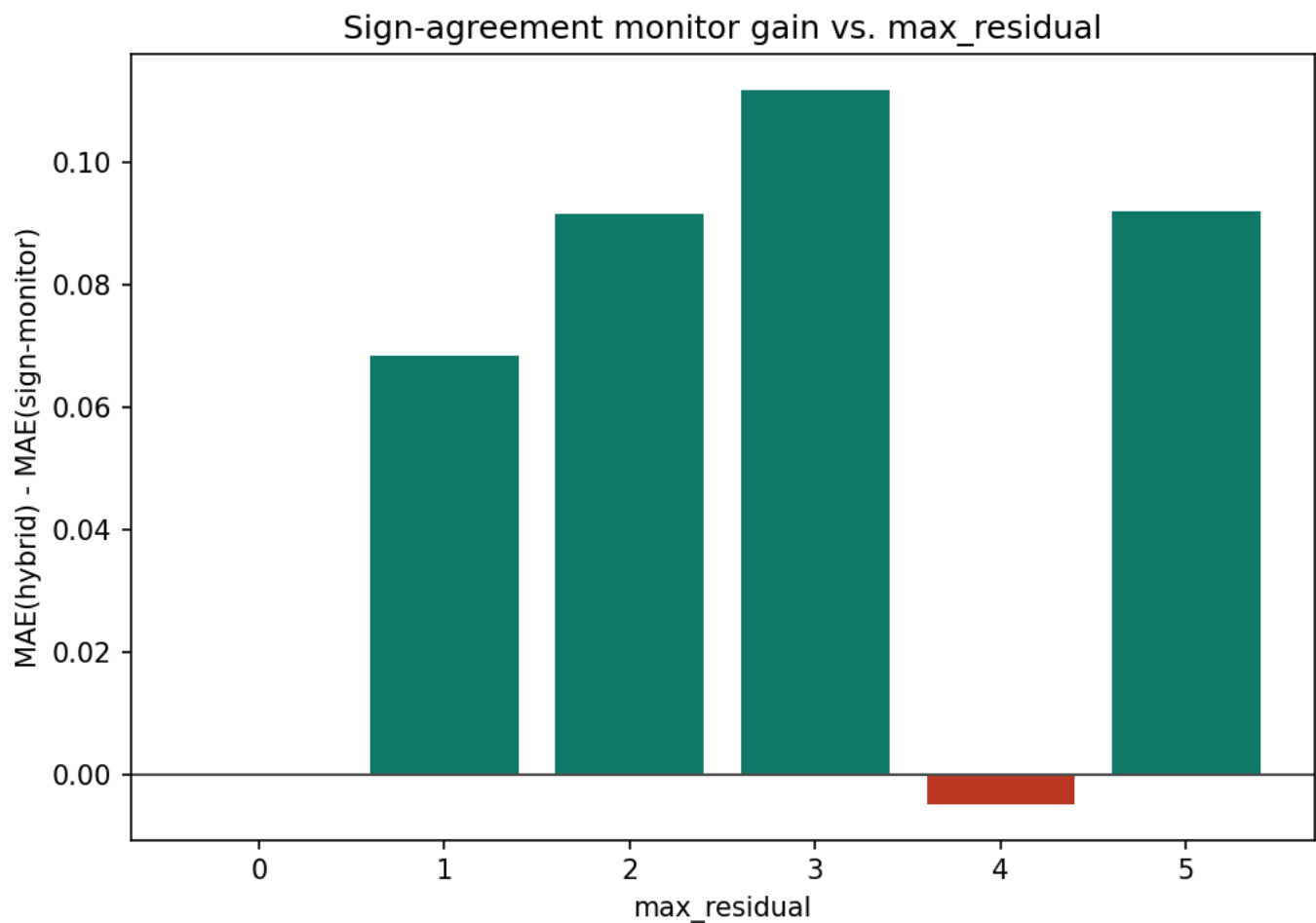
Worst-subject error is flat across $mr=1-4$ (~6.75–6.80 RMSE) then jumps at $mr=5$ (7.003, the sweep's worst) — the only clear sign of a large bound actively hurting the tail.



Once any authority exists, roughly half of all corrections harm the physics baseline and half help — a ratio that is essentially insensitive to how large the bound is.



Counter to intuition: saturation is *highest at the smallest bound* (29% of windows at $mr=1$) and collapses as the bound grows. Since $\rho = |\tanh(\text{raw_delta})|$ does not depend on max_residual mathematically, this shows training itself adapts — a tight bound teaches the network to push raw_delta harder, not the other way around.



The established sign-agreement monitor (re-applied unchanged, no new mechanism) helps at every bound except $mr=4$, where its gain is slightly negative — its value doesn't simply scale with how much authority there is to gate.

WHAT THE SWEEP ANSWERS

There is no clean monotonic tradeoff. $mr=3$ is the best bound by subject win-count (6/11), $mr=4$ the worst (3/11), despite sitting right next to each other on the grid. Subject1's wrong-direction rate holds at a constant 63.6% across every nonzero bound tested — its failure does not depend on how much authority it is given, which is the first direct evidence that the problem is not primarily about the bound.

The learned residual against the ideal one

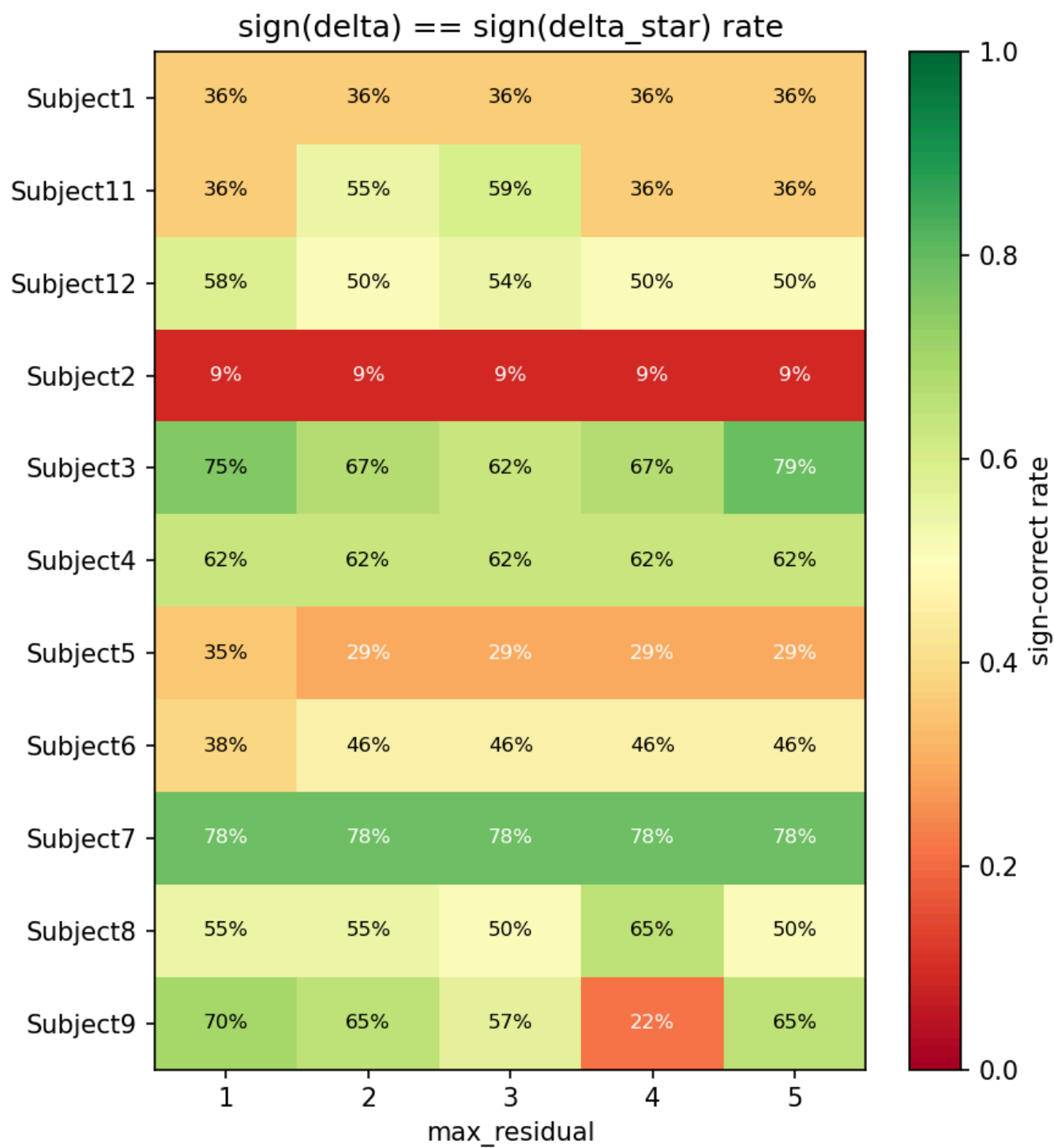
For every window, define `delta_star = y_true - y_linear` — the exact correction that would make the physics baseline perfectly right. Comparing the model's actual `delta` against this ideal, at every bound in {1,2,3,4,5} (mr=0 excluded: its `delta` is identically zero, making the comparison degenerate), answers the question the sweep couldn't: is the correction pointed the right way at all?

Window matching across the five separately-trained runs was verified before drawing any conclusion from it: `y_true` and `y_linear` are bit-identical for the same (`subject_id`, `window_idx`) across all five runs, confirming the windowing and physics calibration never changed — only the model did.

POOLED DELTA VS. DELTA_STAR, BY BOUND

MR	SIGN-CORRECT	DIRECTION FAIL	UNDER-CORRECT	OVER-CORRECT	PEARSON R	MAG. ERROR
1	53.8%	46.2%	43.7%	10.1%	-0.066	-1.746
2	53.3%	46.7%	42.2%	11.1%	+0.023	-1.610
3	52.3%	47.7%	39.7%	12.6%	+0.057	-1.580
4	47.2%	52.8%	35.7%	11.6%	-0.025	-1.676
5	52.3%	47.7%	39.7%	12.6%	+0.018	-1.617

The decisive number is the correlation column. If this were an authority-magnitude problem, `delta` and `delta_star` would correlate strongly and positively, with error concentrated in magnitude. Instead the correlation sits at essentially zero at every bound (-0.07 to +0.06). Sign-correctness hovering at 47–54% — coin-flip — confirms it directly: the network is not reliably choosing the right direction, regardless of how much authority it is given to act on that choice. A smaller secondary pattern exists on top of this — when direction is right, the network under-corrects far more than it over-corrects (magnitude error consistently -1.6 to -1.7) — but it is a minor effect next to the direction problem.

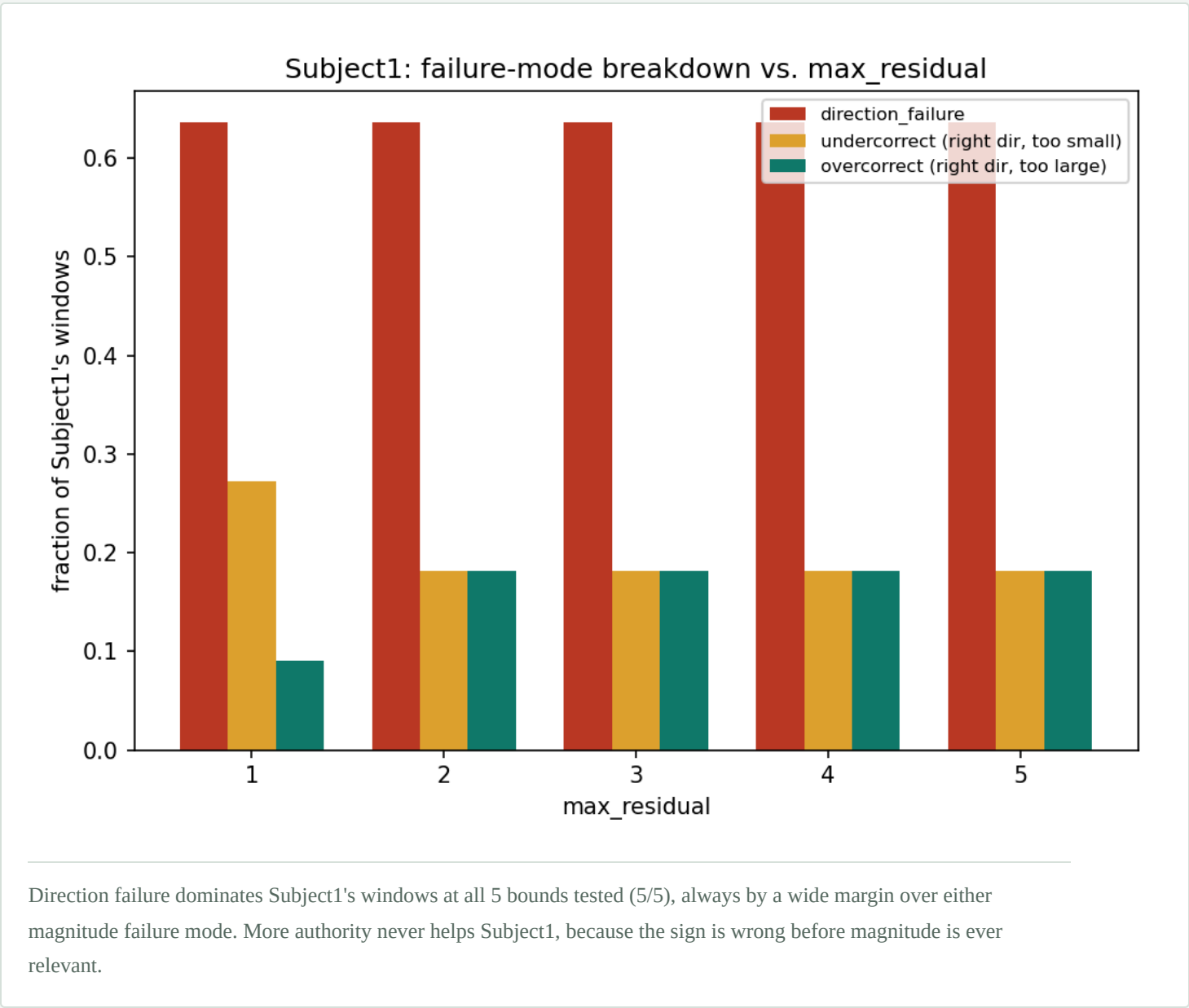


Sign-correctness by subject and bound. Rows are essentially constant across columns — whatever is driving a subject's direction accuracy, it is a property of the subject, not of the bound. Subject2 (9%) is directionally worse than Subject1 (36%) despite not being the subject earlier reports flagged as the worst failure — MAE-based rankings and direction-correctness rankings disagree.

Subject1, diagnosed

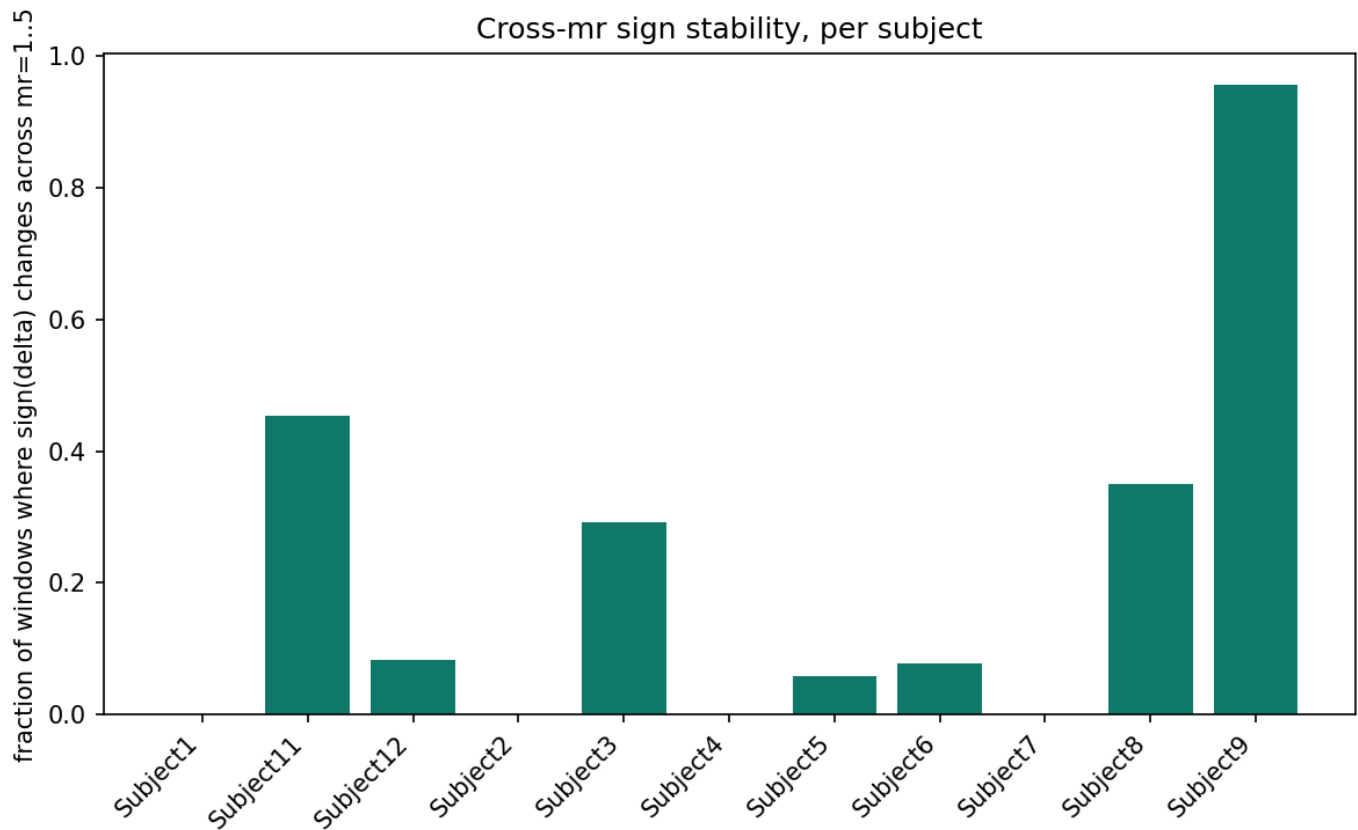
SUBJECT1'S FAILURE-MODE BREAKDOWN, BY BOUND

MR	DIRECTION FAIL	UNDER-CORRECT	OVER-CORRECT
1	63.6%	27.3%	9.1%
2	63.6%	18.2%	18.2%
3	63.6%	18.2%	18.2%
4	63.6%	18.2%	18.2%
5	63.6%	18.2%	18.2%



Is a subject's direction failure reproducible, or noise?

Matching the same physical window across all five independently-trained models tests something the per-bound tables can't: does `sign(delta)` for one window stay the same as the bound changes, or does it flip depending on arbitrary training conditions?



25.1% of windows change sign somewhere across $mr=1..5$, pooled — but this splits subjects into two very different regimes. Subject1, Subject2, Subject4, and Subject7 are perfectly stable (0% change): whatever they get wrong, they get wrong the *same* way every time, regardless of bound. Subject9 is the opposite — 95.7% of its windows change sign depending on the bound, meaning its direction is close to arbitrary noise rather than a systematic miscalibration.

WHY THIS DISTINCTION MATTERS

A pooled "wrong-direction rate" treats Subject1 and Subject9 as the same kind of problem. They are not. Subject1's failure is **reproducible** — a genuine, stable generalization mismatch between its physiology and what the model learned from the other ten subjects, unfixable by adjusting authority. Subject9's failure is closer to **training noise** — sensitive to incidental configuration, which is a different (and potentially more tractable) problem than Subject1's.

Answering the question this report set out to answer

- **The architectural bound is not the lever.** Pooled accuracy is nearly flat across $mr=1-5$; the only large effect is the $0 \rightarrow$ nonzero transition itself, not the specific magnitude chosen within a reasonable range.
- **Saturation behavior runs backwards from intuition** — tight bounds saturate more, not less, because training compensates by pushing the pre-activation harder. This means "shrink the bound to be safer" is not obviously the right lever either.
- **The correlation test is the cleanest evidence in either report so far:** learned and ideal residuals are uncorrelated at every bound tested. This is a direction problem, not a magnitude-calibration problem.
- **Subject1's failure is bound-invariant and reproducible** — the same wrong direction at every authority level, which rules out "more/less authority" as a fix and points at a genuine physiological or calibration mismatch specific to that subject.
- **Not every subject fails the same way.** The sign-stability split (Subject1/2/4/7 stable-but-wrong vs. Subject9 unstable-and-noisy) means a single aggregate diagnosis would have hidden two different problems that likely need two different remedies.

Combined with the previous report's finding that ensemble disagreement provides real but modest, mechanism-uncertain value, the picture across both reports is consistent: the fix that would matter most is not a better bound and not a fancier monitor, but resolving *why* a specific subset of subjects get a stable, wrong answer — which is a generalization question, not a hyperparameter one.